

Magnon Bound States in One-Dimensional Heisenberg Chain

Advanced Solid State Physics - Project Report

Submitted by

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Abstract

This report investigates the formation, spectral signatures, and emergent dynamics of two-particle bound states (magnons) in a one-dimensional interacting fermionic chain, a model mapped from the Heisenberg spin chain via the Jordan-Wigner transformation. Focusing on the strong-coupling regime ($|U| \gg t$), we utilize the two-particle Green's function formalism to establish the exact criteria for bound state formation, demonstrating that a localized “doublon” emerges strictly when the interaction strength exceeds the single-particle kinetic bandwidth ($|U| > 2t$). By analytically calculating the spectral weights and the local density of states, we map this discrete state as it physically separates from the extended two-particle scattering continuum. To understand the effective motion of the bound pair, we employ the Feshbach projection operator method to integrate out high-energy broken-pair configurations. This reduction yields an exact effective tight-binding Hamiltonian, reveal that the tightly bound fermions behave collectively as a single, heavy composite quasiparticle. Overall, the project provides a comprehensive theoretical framework for understanding the emergent excitations and their dynamics in strongly correlated one-dimensional quantum systems.

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1 Introduction

In the study of interacting quantum many-body systems, some of the most striking emergent phenomena arise not from the behavior of individual particles, but from their collective excitations[1]. A paradigm of such emergent behavior is the formation of bound states - composite quasiparticles created when elementary excitations bind together due to strong correlations. Understanding the dynamics of these bound states is a central challenge in modern condensed matter physics, with profound implications for quantum transport, thermalization, and the design of strongly correlated quantum materials[2, 3].

A canonical setting for exploring these phenomena is one-dimensional magnetism, we consider a one-dimensional Heisenberg chain whose Hamiltonian is given by:

$$\hat{H} = -\frac{J}{2} \sum_{i=1}^N (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y) + J_z \sum_{i=1}^N \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^N \sigma_i^z \quad (1)$$

In a magnetic lattice, we observe magnon excitations: quantized spin waves representing propagating spin flips. As first famously demonstrated by Hans Bethe in his exact solution of the 1D Heisenberg chain, strong effective interactions between magnons can overcome their kinetic tendency to delocalize, resulting in the formation of stable multi-magnon bound states[4]. These bound states are fundamentally distinct from independent magnons; they possess their own unique dispersion relations, modified effective masses, and localized relative wavefunctions[5].

To investigate the physics of magnon bound states, it is advantageous to map the spin system onto an equivalent model of fermions or bosons. In one dimension, the Jordan-Wigner transformation provides an exact mapping between spin-1/2 models and chains of interacting spinless fermions[6]. Under this mapping, a local spin flip (a magnon) corresponds to the creation of a fermion, transverse spin exchange corresponds to particle hopping, and longitudinal spin-spin coupling corresponds to nearest-neighbor density-density interactions.

Motivated by this exact equivalence, this report investigates the formation and dynamics of two-particle bound states in a one-dimensional fermionic chain. This minimal yet rich model captures the fundamental competition responsible for bound state formation. The kinetic term drives the delocalization of individual fermions, creating a continuous band of two-particle scattering states. Conversely, the potential term imposes an interaction energy when particles occupy adjacent sites.

We are particularly interested in the strong coupling regime ($|U| \gg t$). In this limit, the massive energy penalty associated with breaking a nearest-neighbor pair drastically alters the scattering dynamics, pulling a localized bound state - often referred to as a doublon - out of the two-particle continuum. Because the constituent fermions obey the Pauli exclusion principle, this composite bound state exhibits nontrivial quantum statistics and constrained kinematics that differ significantly from purely elementary bosons.

In this report, we aim to derive the existence and emergent dynamics of these bound states. We begin by employing the Green's function formalism to identify the spectral poles that correspond to stable two-particle pairing. We then analyze the spectral weights and the local density of states (LDOS) to visualize the separation of the bound state band from the scattering continuum. Finally, utilizing the Feshbach projection operator method, we integrate out the high-energy broken-pair configurations to derive a rigorous effective tight-binding Hamiltonian. This effective model elegantly demonstrates how a tightly bound pair of elementary excitations moves collectively as a single, heavy composite quasiparticle through the lattice.

2 Theoretical Background

2.1 Mapping to Spinless Fermions: Jordan-Wigner Transformation

The Jordan-Wigner transformation is a crucial technique that allows us to exactly map a 1D spin-1/2 chain to a system of interacting spinless fermions[6]. A rigorous definition incorporates a phase factor in the string operator to ensure the correct sign for the fermionic hopping amplitudes. The transformation is defined as:

$$c_j^\dagger = \exp \left[i\pi \sum_i^{j-1} \sigma_i^+ \sigma_i^- \right] \sigma_j^+ ; \quad c_j = \exp \left[-i\pi \sum_i^{j-1} \sigma_i^+ \sigma_i^- \right] \sigma_j^- \quad (2)$$

where $\sigma_i^\pm = \frac{1}{2}(\sigma_i^x \pm i\sigma_i^y)$ are the spin ladder operators. The local fermionic number operator, n_i , is directly related to the z -component of the spin:

$$n_i = c_i^\dagger c_i = \sigma_i^+ \sigma_i^- = \frac{1 + \sigma_i^z}{2} \implies \sigma_i^z = 2c_i^\dagger c_i - 1 = 2n_i - 1 \quad (3)$$

This transformation preserves the fermionic anticommutation relations $\{c_i, c_j^\dagger\} = \delta_{ij}$ and $\{c_i, c_j\} = 0$.

To express the Hamiltonian in Eq. 1 in terms of fermions, we transform each term individually. First, we rewrite the transverse XY interaction using the raising and lowering operators:

$$\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y = 2(\sigma_i^+ \sigma_{i+1}^- + \sigma_i^- \sigma_{i+1}^+) \quad (4)$$

Applying the transformation, the non-local string operators cancel almost entirely, leaving only the z -operator on site i :

$$\sigma_i^+ \sigma_{i+1}^- = c_i^\dagger (-\sigma_i^z) c_{i+1} = c_i^\dagger (1 - 2c_i^\dagger c_i) c_{i+1} = c_i^\dagger c_{i+1} \quad (5)$$

$$\sigma_i^- \sigma_{i+1}^+ = c_i (-\sigma_i^z) c_{i+1}^\dagger = c_i (1 - 2c_i^\dagger c_i) c_{i+1}^\dagger = c_i^\dagger c_i \quad (6)$$

Assuming open boundary conditions, the transverse term maps to standard nearest-neighbor fermionic hopping:

$$-\frac{J}{2} \sum_{i=1}^{N-1} (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y) = -J \sum_{i=1}^{N-1} (c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i) \quad (7)$$

Next, we transform the longitudinal Z interaction (J_z) and the external magnetic field (h) using Eq. 3:

$$\sigma_i^z \sigma_{i+1}^z = (2n_i - 1)(2n_{i+1} - 1) = 4n_i n_{i+1} - 2n_i - 2n_{i+1} + 1 \quad (8)$$

$$-h \sum_{i=1}^N \sigma_i^z = -h \sum_{i=1}^N (2n_i - 1) = -2h \sum_{i=1}^N n_i + hN \quad (9)$$

Summing these contributions and dropping the constant energy shifts (which do not affect the system's eigenstates or dynamics), the fully transformed fermionic Hamiltonian is obtained:

$$\hat{H} = -J \sum_{i=1}^{N-1} (c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i) + V \sum_{i=1}^{N-1} n_i n_{i+1} - \mu \sum_{i=1}^N n_i \quad (10)$$

where $V = 4J_z$ represents a nearest-neighbor density-density interaction, and $\mu = 2h + 4J_z$ acts as an effective chemical potential (approximating for bulk sites). The originally complex Heisenberg XXZ spin chain is thus exactly mapped to a system of interacting 1D spinless fermions.

The derivation above assumes an open chain. If we instead consider a closed ring, the spin Hamiltonian includes a boundary term connecting site N back to site 1. While the longitudinal Z -interaction maps

cleanly across the boundary as $Vn_N n_1$, the transverse XY hopping term behaves differently. Because site N is connected to site 1, the Jordan-Wigner string operator does not cancel out. The boundary hopping term maps to:

$$\sigma_N^+ \sigma_1^- = c_N^\dagger \left(\prod_{j=1}^N (-\sigma_j^z) \right) c_1$$

The unbroken string in the parentheses spans the entire chain and is exactly equivalent to the total fermion number operator, $\exp(i\pi N_f)$, where $N_f = \sum_{i=1}^N n_i$ is the total number of fermions. This means the hopping amplitude across the boundary becomes dependent on the rest of the system: $c_N^\dagger c_1 \rightarrow e^{i\pi N_f} c_N^\dagger c_1$.

Consequently, the boundary conditions for the mapped fermions depend entirely on the total fermion number N_f :

- If N_f is even, $e^{i\pi N_f} = 1$, leading to Anti-Periodic Boundary Conditions ($c_{N+1} = -c_1$).
- If N_f is odd, $e^{i\pi N_f} = -1$, leading to Periodic Boundary Conditions ($c_{N+1} = c_1$).

We will get away with the issue of boundary conditions by working on an infinite lattice. Now that we have mapped the spin system to fermions, we will analyze the fermionic Hamiltonian for the rest of the report, as it provides a more tractable framework for understanding the formation of bound states and their properties.

2.2 One-Dimensional Tight Binding Model

The tight binding model provides a fundamental quantum mechanical description of electrons moving in a periodic potential. We consider a one-dimensional lattice of N sites with a lattice spacing a . Let $c_j^\dagger$ and c_j be the fermionic creation and annihilation operators, respectively, for an electron at site j .

In second quantization, assuming only nearest-neighbor hopping, the Hamiltonian is given by:

$$\hat{H} = -t \sum_{j=1}^N \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) \quad (11)$$

where t is the hopping amplitude between adjacent sites. To impose translational invariance, we apply periodic boundary conditions, imposing $c_{N+1} = c_1$.

2.2.1 Diagonalization by Fourier Transform

Because the system is translationally invariant, the Hamiltonian can be exactly diagonalized by transforming the real-space operators into momentum-space operators using a discrete Fourier transform[7]:

$$c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikja} c_k \quad \text{and} \quad c_j^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-ikja} c_k^\dagger \quad (12)$$

Substituting Eq. 12 into the kinetic (hopping) term of Eq. 11, we obtain:

$$-t \sum_{j=1}^N c_j^\dagger c_{j+1} = -t \sum_{j=1}^N \left(\frac{1}{\sqrt{N}} \sum_k e^{-ikja} c_k^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{k'} e^{ik'(j+1)a} c_{k'} \right) = -t \sum_{k,k'} c_k^\dagger c_{k'} e^{ik'a} \left(\frac{1}{N} \sum_{j=1}^N e^{i(k'-k)ja} \right) \quad (13)$$

$$\implies \hat{H} = \sum_k (-te^{ika} - te^{-ika}) c_k^\dagger c_k = \sum_k \epsilon(k) c_k^\dagger c_k \quad (14)$$

This is the diagonal form of the Hamiltonian. The one-dimensional energy band follows the dispersion relation $\epsilon(k) = -2t \cos(ka)$, as illustrated in Fig. 1a.

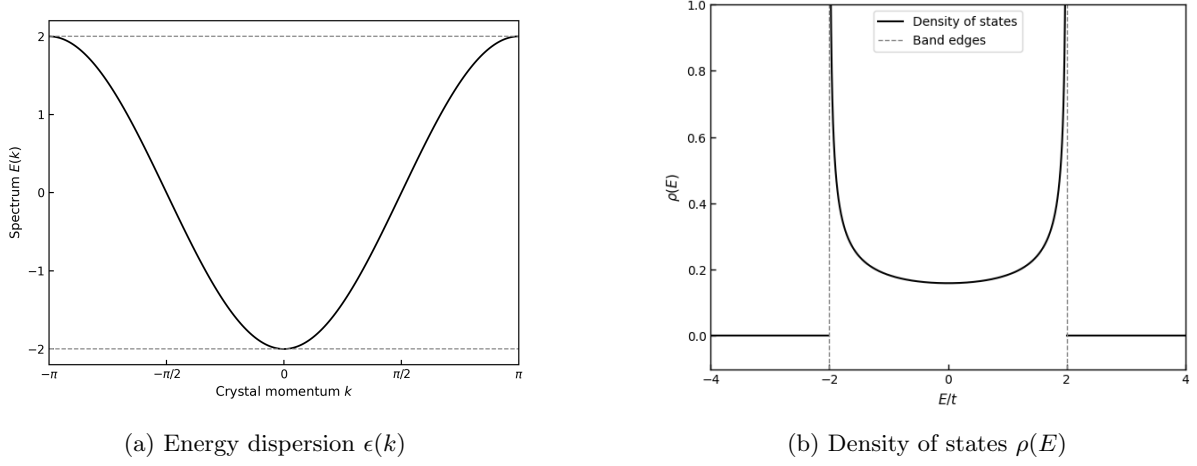


Figure 1: (a) The energy dispersion relation for a 1D tight-binding model with $t = 1, a = 1$. (b) The corresponding density of states, which diverges at the band edges $E = \pm 2t$.

2.2.2 Density of States

The density of states (DOS), denoted as $\rho(E)$, represents the number of available electronic states per unit energy interval. It is formally defined as:

$$\rho(E) = \sum_k \delta(E - \epsilon(k)) \quad (15)$$

In the thermodynamic limit ($N \rightarrow \infty$), the length of the system is $L = Na$, and the discrete sum over momentum k transitions to an integral: $\sum_k \rightarrow \frac{L}{2\pi} \int_{-\pi/a}^{\pi/a} dk$. Including a factor of 2 to account for electron spin degeneracy, the density of states becomes:

$$\rho(E) = 2 \frac{L}{2\pi} \int_{-\pi/a}^{\pi/a} \delta(E - \epsilon(k)) dk \quad (16)$$

To evaluate this integral, we utilize the property of the Dirac delta function $\delta(f(k)) = \sum_i \frac{\delta(k - k_i)}{|f'(k_i)|}$, where k_i are the roots satisfying $E = \epsilon(k)$. First, we compute the derivative of the dispersion relation:

$$\left| \frac{d\epsilon(k)}{dk} \right| = |2ta \sin(ka)| = 2ta \sqrt{1 - \cos^2(ka)} \quad (17)$$

From the dispersion relation, we can isolate the cosine term as $\cos(ka) = -\frac{E}{2t}$. Substituting this into our derivative yields:

$$\left| \frac{d\epsilon(k)}{dk} \right| = 2ta \sqrt{1 - \left(\frac{E}{2t} \right)^2} = a \sqrt{4t^2 - E^2} \quad (18)$$

For any chosen energy E strictly within the band bounds ($|E| < 2t$), there are exactly two roots $\pm k$ in the first Brillouin zone. Summing the equivalent contributions from both roots, the density of states evaluates to:

$$\rho(E) = \frac{2L}{\pi} \frac{1}{\left| \frac{d\epsilon}{dk} \right|} = \frac{2L}{\pi a} \frac{1}{\sqrt{4t^2 - E^2}} \quad (19)$$

The density of states is identically zero for energies outside the band boundaries ($|E| > 2t$) and diverges at the band edges $E = \pm 2t$, as shown in Fig. 1b.

2.3 Two-Dimensional Tight Binding Model on a Square Lattice

We now extend our analysis to a two-dimensional system by considering a square lattice with lattice constant a . The atomic sites are identified by a 2D vector $\mathbf{R}_i = n_x a \hat{x} + n_y a \hat{y}$, where $\mathbf{i} = (n_x, n_y)$ represents the integer coordinates of the lattice sites. To maintain generality, we allow the nearest-neighbor hopping amplitudes to be anisotropic: t_x along the x -direction and t_y along the y -direction.

The second-quantized Hamiltonian with periodic boundary conditions is given by:

$$\hat{H} = - \sum_{\mathbf{i}} \left[t_x \left(c_{\mathbf{i}}^\dagger c_{\mathbf{i}+\hat{x}} + c_{\mathbf{i}+\hat{x}}^\dagger c_{\mathbf{i}} \right) + t_y \left(c_{\mathbf{i}}^\dagger c_{\mathbf{i}+\hat{y}} + c_{\mathbf{i}+\hat{y}}^\dagger c_{\mathbf{i}} \right) \right] \quad (20)$$

where $\mathbf{i} + \hat{x}$ and $\mathbf{i} + \hat{y}$ denote the nearest neighbors to site $\mathbf{i}$ in the x and y directions, respectively.

2.3.1 Diagonalization by Fourier Transform

As in the one-dimensional case, the discrete translational symmetry of the lattice allows us to diagonalize the Hamiltonian by moving to momentum space. We introduce the two-dimensional discrete Fourier transform for the fermionic operators:

$$c_{\mathbf{i}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}_i} c_{\mathbf{k}} \quad \text{and} \quad c_{\mathbf{i}}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}_i} c_{\mathbf{k}}^\dagger \quad (21)$$

where $N = N_x N_y$ is the total number of lattice sites, and the momentum vector $\mathbf{k} = (k_x, k_y)$ is restricted to the first Brillouin zone: $k_x, k_y \in [-\pi/a, \pi/a]$.

Substituting these transformations into the hopping term along the x -direction, we evaluate:

$$\begin{aligned} -t_x \sum_{\mathbf{i}} c_{\mathbf{i}}^\dagger c_{\mathbf{i}+\hat{x}} &= -t_x \sum_{\mathbf{i}} \left(\frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}_i} c_{\mathbf{k}}^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{\mathbf{k}'} e^{i\mathbf{k}' \cdot (\mathbf{R}_i + a\hat{x})} c_{\mathbf{k}'} \right) \\ &= -t_x \sum_{\mathbf{k}, \mathbf{k}'} c_{\mathbf{k}}^\dagger c_{\mathbf{k}'} e^{ik'_x a} \left(\frac{1}{N} \sum_{\mathbf{i}} e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}_i} \right) \end{aligned} \quad (22)$$

The term in the parentheses is the 2D Kronecker delta $\delta_{\mathbf{k}, \mathbf{k}'}$. Applying this logic to all four hopping terms in Eq. 20, the Hamiltonian elegantly decouples into a sum over independent momentum modes:

$$\hat{H} = \sum_{\mathbf{k}} \left(-t_x e^{ik_x a} - t_x e^{-ik_x a} - t_y e^{ik_y a} - t_y e^{-ik_y a} \right) c_{\mathbf{k}}^\dagger c_{\mathbf{k}} = \sum_{\mathbf{k}} \epsilon(\mathbf{k}) c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \quad (23)$$

This yields the two-dimensional dispersion relation:

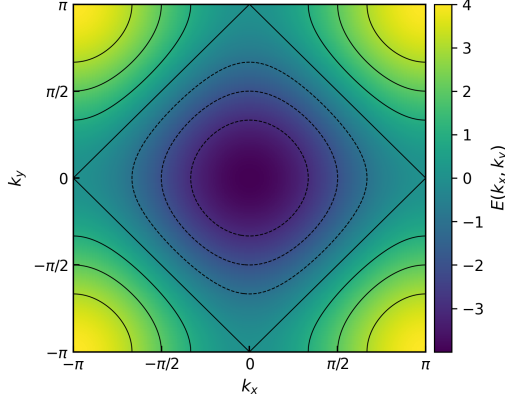
$$\epsilon(k_x, k_y) = -2t_x \cos(k_x a) - 2t_y \cos(k_y a) \quad (24)$$

The total bandwidth of this system is $4|t_x| + 4|t_y|$, and the energy is strictly bounded between $-(2t_x + 2t_y)$ and $+(2t_x + 2t_y)$ (assuming $t_x, t_y > 0$). A density plot of the energy dispersion $\epsilon(k_x, k_y)$ in the first Brillouin zone is shown in Fig. 2a.

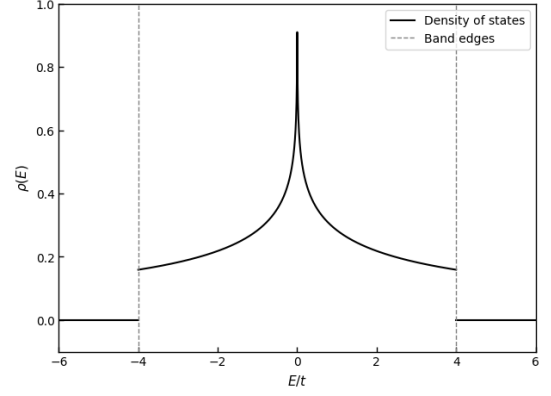
2.3.2 Density of States

We consider the isotropic tight-binding model on a square lattice:

$$\epsilon(k_x, k_y) = -2t [\cos(k_x a) + \cos(k_y a)] \quad (25)$$



(a) Density plot of energy dispersion $\epsilon(k)$



(b) Density of states $\rho(E)$

Figure 2: (a) The energy dispersion relation for a 2D tight-binding model with $t_x = t_y = 1, a = 1$. (b) The corresponding density of states.

The density of states (including spin degeneracy) is:

$$\rho(E) = 2 \frac{L^2}{(2\pi)^2} \int_{\text{BZ}} \delta(E - \epsilon(\mathbf{k})) d^2k \quad (26)$$

Substituting the dispersion relation:

$$\rho(E) = 2 \frac{L^2}{(2\pi)^2} \int_{-\pi/a}^{\pi/a} \int_{-\pi/a}^{\pi/a} \delta(E + 2t(\cos(k_x a) + \cos(k_y a))) dk_x dk_y \quad (27)$$

Using fourfold symmetry of the Brillouin zone:

$$\rho(E) = \frac{2L^2}{\pi^2} \int_0^{\pi/a} \int_0^{\pi/a} \delta(E + 2t(\cos(k_x a) + \cos(k_y a))) dk_x dk_y \quad (28)$$

We now perform the change of variables:

$$u = \cos(k_x a), \quad v = \cos(k_y a) \implies dk_x dk_y = \frac{1}{a^2} \frac{du dv}{\sqrt{1-u^2}\sqrt{1-v^2}} \quad (29)$$

Thus,

$$\rho(E) = \frac{2L^2}{\pi^2 a^2} \int_{-1}^1 \int_{-1}^1 \frac{du dv}{\sqrt{1-u^2}\sqrt{1-v^2}} \delta(E + 2t(u+v)) \quad (30)$$

Using the identity for the Dirac delta function:

$$\delta(E + 2t(u+v)) = \frac{1}{2|t|} \delta\left(v + u + \frac{E}{2t}\right) \quad (31)$$

we integrate over v :

$$\rho(E) = \frac{L^2}{\pi^2 a^2 |t|} \int \frac{du}{\sqrt{1-u^2}\sqrt{1-(u+E/2t)^2}} \quad (32)$$

In the above step, we have used the fact that the delta function enforces $v = -u - \frac{E}{2t}$, which is only valid when $-1 < -u - \frac{E}{2t} < 1$. This condition restricts the limits of integration for u to $[-1, 1] \cap [-1 - E/2t, 1 - E/2t]$. The form of this integral is recognized as the complete elliptic integral of the first kind, $K(z)$, with the argument $z = 1 - \left(\frac{E}{4t}\right)^2$:

$$\rho(E) = \frac{L^2}{\pi^2 a^2 |t|} K \left(1 - \left(\frac{E}{4t} \right)^2 \right), \quad |E| \leq 4|t| \quad (33)$$

and

$$\rho(E) = 0, \quad |E| > 4|t| \quad (34)$$

The most striking feature of this result is its behavior at the center of the band, $E = 0$. As the argument of the elliptic integral approaches 1, $K(z)$ diverges logarithmically.

3 Interacting Fermions in One Dimension

Having established the single-particle properties of the tight-binding model in one and two dimensions, we now shift our focus to the case of two interacting fermions on a one-dimensional tight-binding lattice. We model the interaction as a nearest-neighbor density-density interaction of strength U - this is motivated by the mapping from the Heisenberg spin chain -

$$\hat{H} = -t \sum_{i=1}^{N-1} \left(c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i \right) + U \sum_{i=1}^{N-1} n_i n_{i+1} \quad (35)$$

3.1 Density of States of the Non-Interacting Hamiltonian

Before we delve into the effects of interactions, it is instructive to first understand the two-particle density of states for the non-interacting part of the Hamiltonian:

$$\rho_2(E) = \int dE_1 dE_2 \rho_1(E_1) \rho_1(E_2) \delta(E - E_1 - E_2) = \int dE_1 \rho_1(E_1) \rho_1(E - E_1) \quad (36)$$

where $\rho_1(E) = \frac{2L}{\pi a} \frac{1}{\sqrt{4t^2 - E^2}}$ is the single-particle density of states. Substituting this into the expression for $\rho_2(E)$, we find:

$$\rho_2(E) = \frac{4L^2}{\pi^2 a^2} \int \frac{dE_1}{\sqrt{4t^2 - E_1^2} \sqrt{4t^2 - (E - E_1)^2}} \quad (37)$$

$$= \frac{L^2}{\pi^2 a^2 t^2} \int \frac{dE_1}{\sqrt{1 - \left(\frac{E_1}{2t}\right)^2} \sqrt{1 - \left(\frac{E - E_1}{2t}\right)^2}} \quad (38)$$

$$= \frac{2L^2}{\pi^2 a^2 t} \int \frac{dx}{\sqrt{1 - x^2} \sqrt{1 - \left(\frac{E}{2t} - x\right)^2}} = \frac{2L^2}{\pi^2 a^2 t} K \left(1 - \left(\frac{E}{4t} \right)^2 \right) \quad (39)$$

Interestingly, the two-particle density of states in one dimension has the same functional form as the single-particle density of states in two dimensions.

3.2 Two-particle basis

By virtue of translational invariance, the center-of-mass momentum k is a good quantum number, and let us define the state $|k, n\rangle$ as the two-particle state with total momentum k and relative separation n between the two fermions:

$$|k, n\rangle = \frac{1}{\sqrt{N}} \sum_j \exp \left(ik \left(R_j + \frac{na}{2} \right) \right) c_j^\dagger c_{j+n}^\dagger |0\rangle \quad (40)$$

where $R_j = ja$ is the position of the j -th lattice site. The states $\{|k, n\rangle\}$ form an orthonormal basis for the two-particle sector of the Hilbert space. The orthonormality can be verified by direct calculation:

$$\begin{aligned}
\langle k', n' | k, n \rangle &= \frac{1}{N} \sum_{j, j'} \exp \left[-ik' \left(R_{j'} + \frac{n'a}{2} \right) + ik \left(R_j + \frac{na}{2} \right) \right] \langle 0 | c_{j'+n'} c_{j'} c_j^\dagger c_{j+n}^\dagger | 0 \rangle \\
&= \frac{1}{N} \sum_{j, j'} \exp \left[-ik' a \left(j' + \frac{n'}{2} \right) + ika \left(j + \frac{n}{2} \right) \right] \delta_{j, j'} \delta_{n, n'} \\
&= \delta_{n, n'} \frac{1}{N} \sum_j \exp \left[i(k - k') a \left(j + \frac{n}{2} \right) \right] \\
&= \delta_{n, n'} \exp \left[i(k - k') \frac{na}{2} \right] \left(\frac{1}{N} \sum_j \exp [i(k - k') ja] \right) \\
&= \delta_{n, n'} \exp \left[i(k - k') \frac{na}{2} \right] \delta_{k, k'} \\
&= \delta_{n, n'} \delta_{k, k'}
\end{aligned}$$

The completeness of the $\{|k, n\rangle\}$ basis follows directly from the completeness of the real-space two-particle basis. The $\{|k, n\rangle\}$ states are simply obtained by performing a unitary transform over the center-of-mass coordinate of the real-space states, which is an invertible operation that preserves completeness.

4 The Resolvent Operator and Green's Function

4.1 Definitions and Lehmann Representation

The resolvent operator is defined as $\hat{G}(\omega) = (\omega + i\eta - \hat{H})^{-1}$, where $\eta \rightarrow 0^+$ is a positive infinitesimal. The inclusion of $i\eta$ serves a dual purpose. Mathematically, it acts as a regularization parameter that shifts the poles of the operator slightly into the lower half of the complex energy plane, avoiding singularities at the eigenvalues of the Hamiltonian $\hat{H}$. Physically, it enforces causality; this specific pole structure ensures that the Fourier transform to the time domain yields the *retarded* Green's function, meaning the system's response only occurs strictly after a perturbation. When interactions are introduced, η is effectively replaced by a finite contribution from self-energy term, Γ . In the time domain, this introduces an exponentially decaying factor to the probability amplitude ($e^{-\Gamma t}$), which corresponds to a finite quasiparticle lifetime of $\tau = 1/(\Gamma)$. In essence, the resolvent operator encodes the full spectrum of the Hamiltonian and its response to perturbations.

The two-particle Green's function in the $|k, n\rangle$ basis is defined as:

$$G(m, n; k, \omega) = \langle k, m | \hat{G}(\omega) | k, n \rangle \quad (41)$$

This function describes the amplitude for two fermions, initially separated by n lattice sites, to propagate and be found separated by m lattice sites at energy ω [8]. The poles of $G(m, n; k, \omega)$ correspond to the energy eigenvalues of the two-particle system. Let us denote the two-particle eigenstates of $\hat{H}$ as $|k, \alpha\rangle$ with corresponding eigenvalues E_α . We can express the Green's function in terms of these eigenstates:

$$\hat{H} = \sum_\alpha E_\alpha |k, \alpha\rangle \langle k, \alpha| \quad \implies \quad \hat{G}(\omega) = \sum_\alpha \frac{|k, m\rangle \langle k, \alpha| \langle k, \alpha| k, n\rangle}{\omega + i\eta - E_\alpha} \quad (42)$$

This is referred to as the Lehmann representation of the Green's function. If we perform the Fourier transform to the time domain, we obtain $G(m, n; k, t) \propto \langle k, m | e^{-i\hat{H}t} | k, n \rangle$, which describes overlap of the time-evolved two particle state $|k, n\rangle$ with the state $|k, m\rangle$.

4.2 Solution of the Two-Particle Problem

4.2.1 Setting up the Recurrence Relation

To solve for the two-particle Green's function, we set up a recurrence relation based on the action of the Hamiltonian on the $|k, n\rangle$ states. By definition,

$$\mathbb{I} = \hat{G}(\omega)(\omega + i\eta - \hat{H}) \implies \langle k, m | \mathbb{I} | k, n \rangle = (\omega + i\eta) \langle k, m | \hat{G}(\omega) | k, n \rangle - \langle k, m | \hat{G}(\omega) \hat{H} | k, n \rangle \quad (43)$$

$$\implies \delta_{m,n} = (\omega + i\eta) G(m, n; k, \omega) - \langle k, m | \hat{G}(\omega) \hat{H} | k, n \rangle \quad (44)$$

In the above step, we have used the orthonormality of the $|k, n\rangle$ states and the definition of the Green's function. Further, note that -

$$\begin{aligned} \hat{H} |k, n\rangle &= \left[-t \sum_{j'} (c_{j'}^\dagger c_{j'+1} + c_{j'+1}^\dagger c_{j'}) + U \sum_{j'} n_{j'} n_{j'+1} \right] \frac{1}{\sqrt{N}} \sum_j \exp\left(ika \left(j + \frac{n}{2}\right)\right) c_j^\dagger c_{j+n}^\dagger |0\rangle \\ &= \frac{-t}{\sqrt{N}} \sum_j \exp\left(ika \left(j + \frac{n}{2}\right)\right) (c_{j+1}^\dagger c_{j+n}^\dagger + c_{j-1}^\dagger c_{j+n}^\dagger + c_j^\dagger c_{j+n+1}^\dagger + c_j^\dagger c_{j+n-1}^\dagger) |0\rangle \\ &\quad + \frac{U}{\sqrt{N}} \sum_j \exp\left(ika \left(j + \frac{n}{2}\right)\right) \left(\sum_{j'} n_{j'} n_{j'+1} c_j^\dagger c_{j+n}^\dagger |0\rangle \right) \\ &= \frac{-t}{\sqrt{N}} \sum_j \left[\exp\left(ika \left(j + \frac{n}{2}\right)\right) c_{j+1}^\dagger c_{j+n}^\dagger + \exp\left(ika \left(j + \frac{n}{2}\right)\right) c_j^\dagger c_{j+n+1}^\dagger \right. \\ &\quad \left. + \exp\left(ika \left(j + \frac{n}{2}\right)\right) c_{j-1}^\dagger c_{j+n}^\dagger + \exp\left(ika \left(j + \frac{n}{2}\right)\right) c_j^\dagger c_{j+n-1}^\dagger \right] |0\rangle + U \delta_{n,1} |k, n\rangle \\ &= -t \left(\exp\left(-i\frac{ka}{2}\right) |k, n-1\rangle + \exp\left(-i\frac{ka}{2}\right) |k, n+1\rangle + \exp\left(i\frac{ka}{2}\right) |k, n+1\rangle + \exp\left(i\frac{ka}{2}\right) |k, n-1\rangle \right) \\ &\quad + U \delta_{n,1} |k, n\rangle \\ &= -t \left(\exp\left(i\frac{ka}{2}\right) + \exp\left(-i\frac{ka}{2}\right) \right) (|k, n-1\rangle + |k, n+1\rangle) + U \delta_{n,1} |k, n\rangle \\ &= -2t \cos\left(\frac{ka}{2}\right) (|k, n-1\rangle + |k, n+1\rangle) + U \delta_{n,1} |k, n\rangle \end{aligned}$$

Hence, it follows that:

$$\langle k, m | \hat{G}(\omega) \hat{H} | k, n \rangle = -f(k) (G(m, n-1; k, \omega) + G(m, n+1; k, \omega)) + U \delta_{n,1} G(m, n; k, \omega) \quad (45)$$

$$\implies \delta_{m,n} = (\omega + i\eta - U \delta_{n,1}) G(m, n; k, \omega) + f(k) (G(m, n-1; k, \omega) + G(m, n+1; k, \omega)) \quad (46)$$

where, $f(k) = 2t \cos\left(\frac{ka}{2}\right)$ can be thought of as an effective hopping amplitude for the relative motion of the two particles. Let us define $a_{m,n} = G(m, n; k, \omega)$ for brevity. Then, the above equation can be recasted as a recurrence relation:

$$\delta_{m,n} = (\omega + i\eta - U \delta_{n,1}) a_{m,n} + f(k) (a_{m,n-1} + a_{m,n+1}) \quad (47)$$

This is a second-order linear difference equation in two variables. By fixing m and treating n as the variable, we can solve this equation for each fixed m . This follows from the fact that $G(m, n; k, \omega) \rightarrow 0$ in the limit of $n \rightarrow \infty$ for any fixed m . The requirement is a necessary boundary condition reflecting the physical normalizability of the two-particle states. For energies ω outside the two-particle scattering continuum, the states are fundamentally bound, meaning the relative wavefunction is highly localized and must decay exponentially with increasing separation n . For energies within the scattering continuum, the asymptotic behavior is dictated by the infinitesimal imaginary part $+i\eta$ which enforces an exponentially

damped outgoing-wave boundary condition, ensuring that the transition amplitude vanishes at infinite separation regardless of the specific spectral regime. Since the real-space Green's function is constructed as a sum over all energy eigenstates as given by the Lehmann representation, the Green's function is directly proportional to $\langle k, \alpha | k, n \rangle$, if every individual state $|k, \alpha\rangle$ vanishes at spatial infinity, the Green's function also vanishes.

4.2.2 Solution of the Green's function for $m=1$

Let us use the notation $a_n = a_{1,n} = G(1, n; k, \omega)$, then the recurrence relation for $m = 1$ is:

$$(\omega + i\eta - U)a_1 + f(k)a_2 = 1 \quad (48)$$

$$(\omega + i\eta)a_n + f(k)(a_{n-1} + a_{n+1}) = 0, \quad n \geq 2 \quad (49)$$

For $ka = \pi$, $f(k) = 0$, this makes the solution trivial: $a_1 = \frac{1}{\omega + i\eta - U}$ and $a_n = 0$ for $n \geq 2$. For $ka \neq \pi$, let us make the ansatz $a_n = \alpha z^n$ for $n \geq 2$. Substituting this into the second equation, we find:

$$(\omega + i\eta)\alpha z^n + f(k)\alpha(z^{n-1} + z^{n+1}) = 0 \implies (\omega + i\eta) + f(k)\left(z + \frac{1}{z}\right) = 0 \quad (50)$$

$$\implies z = \frac{1}{2f(k)} \left(-(\omega + i\eta) \pm \sqrt{(\omega + i\eta)^2 - 4f(k)^2} \right) = \frac{1}{2} \left(\frac{-(\omega + i\eta)}{f(k)} \pm \sqrt{\left(\frac{\omega + i\eta}{f(k)}\right)^2 - 4} \right) \quad (51)$$

The two roots of the characteristic equation are reciprocals of each other, $z_1 z_2 = 1$. Let z be the root with magnitude less than 1, then the general solution to the recurrence relation is $a_n = \alpha z^n + \beta z^{-n}$. Since $a_n \rightarrow 0$ as $n \rightarrow \infty$ in the thermodynamic limit, we impose $\beta = 0$. Substituting the solution back into the first equation, we find:

$$a_1 = \frac{1 - f(k)\alpha z^2}{\omega + i\eta - U} \quad (52)$$

Substituting this into the equation for a_2 , we find:

$$(\omega + i\eta)\alpha z^2 + f(k)\left(\frac{1 - f(k)\alpha z^2}{\omega + i\eta - U} + \alpha z^3\right) = 0 \quad (53)$$

$$\implies \alpha \left((\omega + i\eta)z^2 + f(k)z^3 - \frac{f(k)^2 z^2}{\omega + i\eta - U} \right) = -\frac{f(k)}{\omega + i\eta - U} \quad (54)$$

$$\implies \alpha = -\left((\omega + i\eta)z^2 + f(k)z^3 - \frac{f(k)^2 z^2}{\omega + i\eta - U} \right)^{-1} \frac{f(k)}{\omega + i\eta - U} \quad (55)$$

Using the characteristic equation, $(\omega + i\eta)z^2 + f(k)z^3 = -f(k)z$, we can simplify the expression for α :

$$\alpha = -\left(-f(k)z - \frac{f(k)^2 z^2}{\omega + i\eta - U} \right)^{-1} \frac{f(k)}{\omega + i\eta - U} = \frac{1}{(\omega + i\eta - U)z + f(k)z^2} = \frac{1}{-Uz - f(k)} = -\frac{1}{Uz + f(k)} \quad (56)$$

Finally, the solution for the Green's function is:

$$a_1 = \frac{1}{\omega + i\eta - U} \left(1 + \frac{f(k)z^2}{Uz + f(k)} \right) = \left(\frac{z}{-f(k)z^2 - f(k) - Uz} \right) \left(\frac{Uz + f(k) + f(k)z^2}{Uz + f(k)} \right) \quad (57)$$

$$= -\frac{1}{U + \frac{f(k)}{z}} = -\frac{1}{U - \omega - i\eta - f(k)z} = (\omega + i\eta - U + f(k)z)^{-1} \quad (58)$$

$$a_n = -\frac{z^n}{Uz + f(k)} = -\frac{z^{n-1}}{U + \frac{f(k)}{z}} = z^{n-1} a_1; \quad \forall n \geq 2 \quad (59)$$

5 Bound States in the Fermionic Chain: Existence and Spectral Signatures

In this section, we analyze the existence of two-particle bound states in the fermionic chain. Two-fermion bound states correspond to poles of the Green's function that lie outside the two-particle scattering continuum. In real space, these states are characterized by an exponentially decaying two-particle wavefunction, with the density of the two particles being highest when they are in the nearest-neighbor configuration ($n = 1$).

5.1 Spectral Weights

To mathematically determine the existence of bound states, we study $G(1, 1; k, \omega) = \langle k, 1 | \hat{G}(\omega) | k, 1 \rangle$ - which encodes the amplitude for two particles, initially separated by one lattice site, to propagate and be found separated by one lattice site at energy ω . The imaginary part of $G(1, 1; k, \omega)$ is directly related to the spectral weight of the two-particle states $A_2(k, \omega)$ at momentum k and energy ω [9]:

$$A_2(k, \omega) = -\frac{1}{\pi} \text{Im } G(1, 1; k, \omega) \quad (60)$$

The spectral weight $A_2(k, \omega)$ is a measure of the density of two-particle states at a given momentum and energy, specifically projected onto the nearest-neighbor configuration. A bound state manifests as a sharp peak in $A_2(k, \omega)$ that lies outside the two-particle scattering continuum. Integrating $A_2(k, \omega)$ over the entire Brillouin zone yields the density of two-particle states for this specific $n = 1$ separation.

5.1.1 Analytical form as $\eta \rightarrow 0^+$

In this section, we will derive the explicit form of $A_2(k, \omega)$ in the limit of $\eta \rightarrow 0^+$ using the solution for $G(1, 1; k, \omega)$ obtained in the previous section, $G(1, 1; k, \omega) = (\omega + i\eta - U + f(k)z)^{-1}$, where z is the root of the characteristic equation with magnitude less than 1.

1. The Scattering Continuum: For energies $|\omega| < 2|f(k)|$, the system supports extended scattering states. The roots of the characteristic equation are complex with $|z| = 1$, taking the form $f(k)z = -\omega/2 + i\sqrt{4f(k)^2 - \omega^2}/2$. Substituting this into the Green's function, we explicitly rationalize the denominator to extract the imaginary part:

$$\begin{aligned} \text{Im } G(1, 1; k, \omega) &= \text{Im} \left(\frac{1}{\omega - U + \left(-\frac{\omega}{2} + i\frac{\sqrt{4f(k)^2 - \omega^2}}{2} \right)} \right) \\ &= \text{Im} \left(\frac{2}{\omega - 2U + i\sqrt{4f(k)^2 - \omega^2}} \cdot \frac{\omega - 2U - i\sqrt{4f(k)^2 - \omega^2}}{\omega - 2U - i\sqrt{4f(k)^2 - \omega^2}} \right) \\ &= \frac{-2\sqrt{4f(k)^2 - \omega^2}}{\omega^2 - 4U\omega + 4U^2 + 4f(k)^2 - \omega^2} \\ &= \frac{-2\sqrt{4f(k)^2 - \omega^2}}{4U^2 - 4U\omega + 4f(k)^2} = -\frac{\sqrt{4f(k)^2 - \omega^2}}{2(U^2 - U\omega + f(k)^2)} \end{aligned} \quad (61)$$

2. The Bound State Pole: For energies outside the continuum ($|\omega| > 2|f(k)|$), z is purely real. The imaginary part of the Green's function vanishes everywhere except at the isolated poles where the denominator $D(\omega) = \omega - U + f(k)z(\omega)$ crosses zero. Using the characteristic energy relation $\omega =$

$-f(k)(z + z^{-1})$, the pole condition mandates:

$$-f(k) \left(z + \frac{1}{z} \right) - U + f(k)z = 0 \quad \implies \quad -\frac{f(k)}{z} - U = 0 \quad \implies \quad z_p = -\frac{f(k)}{U} \quad (62)$$

Because a normalizable bound state requires an exponentially decaying relative wavefunction ($|z_p| < 1$), this state only emerges when: $|U| > |f(k)|$. The exact energy of this localized pair is given by evaluating ω at the pole, yielding $\omega_{BS} = U + f(k)^2/U$.

To find the spectral weight Z contained in this discrete delta-function peak, we must calculate the residue of the Green's function at the pole[9]. Since the pole is a simple root of the denominator, the residue is $Z = 1/D'(\omega_{BS})$. We evaluate the derivative using the chain rule:

$$D'(\omega) = \frac{d}{d\omega} \left(\omega - U + f(k)z(\omega) \right) = 1 + f(k) \frac{dz}{d\omega} \quad (63)$$

To find $dz/d\omega$, we differentiate the characteristic relation $\omega = -f(k)(z + z^{-1})$ with respect to z :

$$\frac{d\omega}{dz} = -f(k) \left(1 - \frac{1}{z^2} \right) = -f(k) \left(\frac{z^2 - 1}{z^2} \right) \quad \implies \quad \frac{dz}{d\omega} = -\frac{z^2}{f(k)(z^2 - 1)} \quad (64)$$

Substituting this back into $D'(\omega)$ and evaluating at the pole z_p :

$$D'(\omega_{BS}) = 1 + f(k) \left(-\frac{z_p^2}{f(k)(z_p^2 - 1)} \right) = 1 - \frac{z_p^2}{z_p^2 - 1} = \frac{1}{1 - z_p^2} = 1 - \frac{f(k)^2}{U^2} \quad (65)$$

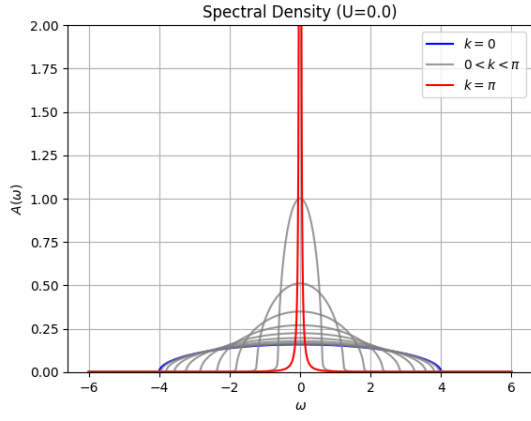
Full Spectral Weight: Combining both regimes, the exact analytical form of the projected two-particle spectral weight is:

$$A_2(k, \omega) = \begin{cases} \frac{1}{2\pi} \frac{\sqrt{4f(k)^2 - \omega^2}}{U^2 - U\omega + f(k)^2} & \text{for } |\omega| < 2|f(k)| \\ \left(1 - \frac{f(k)^2}{U^2} \right) \delta \left(\omega - U - \frac{f(k)^2}{U} \right) & \text{for } |\omega| > 2|f(k)| \text{ and } |U| > |f(k)| \\ 0 & \text{otherwise} \end{cases} \quad (66)$$

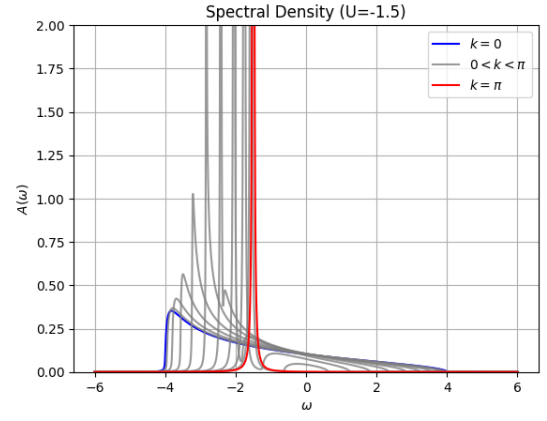
5.1.2 Spectral weights at finite η

Unlike the $\eta \rightarrow 0^+$ limit, the analytical form of $A_2(k, \omega)$ at finite η is more complicated and does not admit a simple closed-form expression. Here, we present few plots of $A_2(k, \omega)$ at finite $\eta = 0.01$ for various values of U and k . The plots are generated using the exact solution for $G(1, 1; k, \omega)$ at finite η .

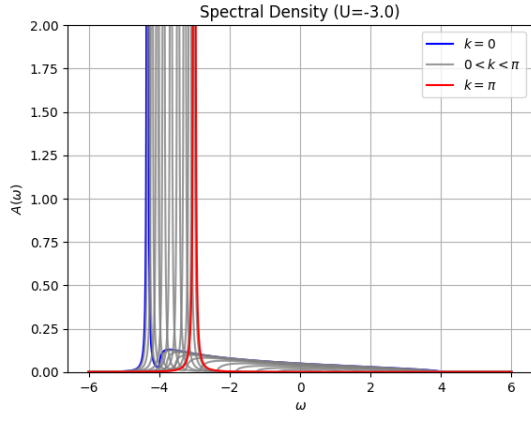
When the interaction strength is zero ($U = 0$), $A_2(k, \omega)$ is nonzero only within the scattering continuum $[-4t, 4t]$ and vanishes outside. At $ka = \pi$, the spectral weight diverges at $\omega = 0$, implying that $\omega = 0$ corresponds to a two-particle nearest-neighbor bound state. As the interaction strength increases, the $ka = \pi$ peak always happens at $\omega = U$, as seen in figure 3a. This can be explained by the fact that $G(1, 1; \frac{\pi}{a}, \omega) = (\omega + i\eta - U)^{-1}$, which has a pole at $\omega = U$ for any value of U . Even for ka near to π , we find that as we increase U , a sharp peak emerges towards the ends of scattering continuum. Decreasing U beyond $-2t$, the bound state peak moves further away from the continuum limit and becomes the lower-energy state for all k , as seen in figure 3c. This is also the case when the interaction is repulsive and $U > 2t$, leading to a repulsively bound state above the scattering continuum, as seen in figure 3d. For $U \gg t$, the bound state is well separated from the scattering continuum and corresponds to a large energy cost for breaking the pair, which leads to a very sharp peak in the spectral weight and an increased quasiparticle lifetime.



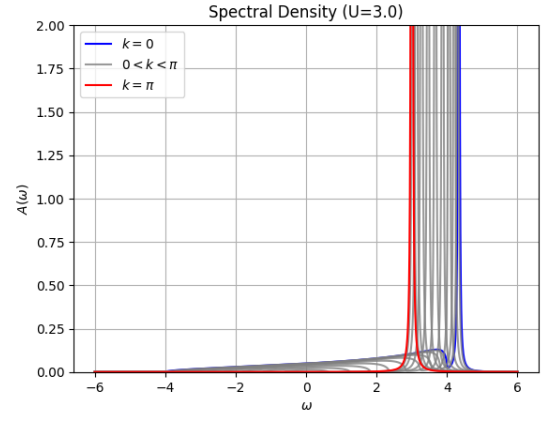
(a) $U = 0$



(b) $U = -1.5$



(c) $U = -3$



(d) $U = 3$

Figure 3: Spectral weight $A_2(k, \omega)$ at finite $\eta = 0.01$ for $U = 0, -1.5, \pm 3$, $t = 1$, and different values of k .

5.2 Local Density of States

Having found out the spectral weight for different k , the full density of states for the nearest-neighbor two-particle sector can be calculated by $\rho(\omega) = \sum_{\text{BZ}} A_2(k, \omega)$. This has been done numerically by considering a finite lattice with $N = 100$ sites and summing over all discrete k values in the Brillouin zone. The resulting density of states is plotted below for different values of U .

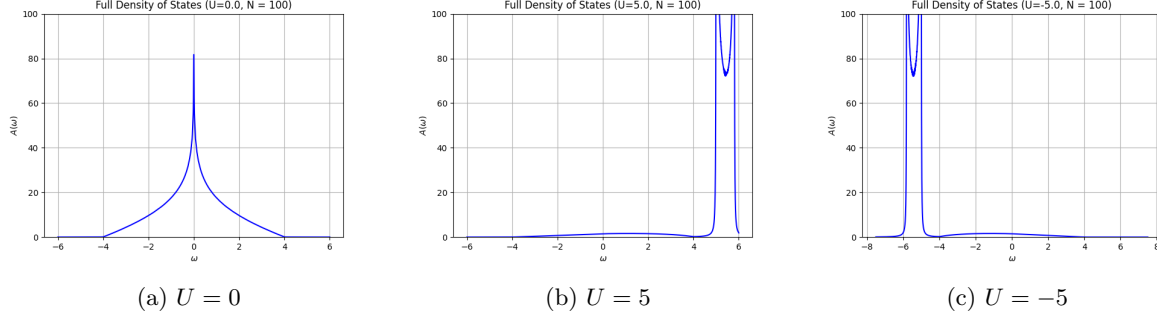


Figure 4: Density of states $\rho(\omega)$ for the nearest-neighbor two-particle sector at finite $\eta = 0.01$ for $U = 0, \pm 5$, $t = 1$, and $N = 100$ sites.

For $U = 0$, we find that the nn sector density of states is nonzero only within the scattering continuum $[-4t, 4t]$ and resembles the full density of states for a two-particle system without interactions. This is not a mere coincidence, but can be explained as follows.

For $U = 0$, the projected spectral weight is $A_2(k, \omega) = \frac{1}{2\pi} \frac{\sqrt{4f(k)^2 - \omega^2}}{f(k)^2}$. Integrating this over the Brillouin zone yields the local density of states. Because the spectral weight is only real when $4f(k)^2 \geq \omega^2$, we restrict our integral to the momentum cutoff k_c where $f(k_c) = |\omega|/2$. Exploiting the even symmetry of the integrand, we have:

$$\rho_{n=1}(\omega) = 2 \int_0^{k_c} \frac{dk}{2\pi} \frac{1}{2\pi} \frac{\sqrt{4f(k)^2 - \omega^2}}{f(k)^2} = \frac{1}{2\pi^2} \int_0^{k_c} dk \frac{\sqrt{4f(k)^2 - \omega^2}}{f(k)^2} \quad (67)$$

Let us make the change of variable $x = f(k) = 2t \cos(\frac{ka}{2})$. The differential is $dx = -ta \sin(\frac{ka}{2}) dk = -\frac{a}{2} \sqrt{4t^2 - x^2} dk$. At the boundaries, $k = 0 \Rightarrow x = 2t$, and $k = k_c \Rightarrow x = |\omega|/2$. Substituting these into the integral gives:

$$\rho_{n=1}(\omega) = \frac{1}{2\pi^2} \int_{|\omega|/2}^{2t} \left(\frac{2 dx}{a \sqrt{4t^2 - x^2}} \right) \frac{\sqrt{4x^2 - \omega^2}}{x^2} = \frac{1}{\pi^2 a} \int_{|\omega|/2}^{2t} dx \frac{\sqrt{4x^2 - \omega^2}}{x^2 \sqrt{4t^2 - x^2}} \quad (68)$$

To evaluate this, we use the standard elliptic substitution $x = 2t\sqrt{1 - m \sin^2 \theta}$, where the parameter $m = 1 - (\frac{\omega}{4t})^2$. Under this transformation, the integration limits $|\omega|/2 \rightarrow 2t$ map exactly to $\pi/2 \rightarrow 0$. The differential becomes $dx = \frac{-2tm \sin \theta \cos \theta}{\sqrt{1 - m \sin^2 \theta}} d\theta$, and the integrand components simplify to $\sqrt{4x^2 - \omega^2} = 2t\sqrt{m} \sin \theta$ and $\sqrt{4t^2 - x^2} = 4t\sqrt{m} \cos \theta$. Reversing the limits to absorb the negative sign, we obtain:

$$\begin{aligned} \rho_{n=1}(\omega) &= \frac{1}{\pi^2 a} \int_0^{\pi/2} \left(\frac{2tm \sin \theta \cos \theta}{\sqrt{1 - m \sin^2 \theta}} d\theta \right) \frac{4t\sqrt{m} \cos \theta}{4t^2(1 - m \sin^2 \theta) \cdot 2t\sqrt{m} \sin \theta} \\ &= \frac{1}{\pi^2 a t} \int_0^{\pi/2} \frac{m \cos^2 \theta}{(1 - m \sin^2 \theta)^{3/2}} d\theta \end{aligned} \quad (69)$$

Using the algebraic identity $m \cos^2 \theta = 1 - m \sin^2 \theta - (1 - m)$, we can split this into two standard integrals:

$$\begin{aligned}\rho_{n=1}(\omega) &= \frac{1}{\pi^2 at} \left[\int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - m \sin^2 \theta}} - (1 - m) \int_0^{\pi/2} \frac{d\theta}{(1 - m \sin^2 \theta)^{3/2}} \right] \\ &= \frac{1}{\pi^2 at} [K(m) - E(m)]\end{aligned}\quad (70)$$

where $K(m)$ and $E(m)$ are the Complete Elliptic Integrals of the First and Second Kind, respectively, evaluated at $m = 1 - (\omega/4t)^2$. The analytical form of the local density of states $\rho_{n=1}(\omega)$ is plotted in figure 5, which matches perfectly with the result in figure 4a. Due to the $K(m)$ term, the local density of states behaves like the full two-particle density of states, and the $E(m)$ term leads to the local density of states vanishing at the band edges, as seen in figure 4a.

Also for $|U| > 2t$, one can see that the local density of states is dominated by a narrow band that emerges outside the scattering continuum, which corresponds to the bound state. As seen in figures 4b and 4c, the bound state can either be attractively bound below the continuum for $U < -2t$, or repulsively bound above the continuum for $U > 2t$.

Interestingly, the bound state density of states looks very similar to the density of states for a single-particle tight-binding model shifted by an energy $\approx U$. This is not a coincidence, but has a deeper physical explanation - we will settle this in the next section when we analyze the effective Hamiltonian for the bound state in the strong coupling limit $|U| \gg t$.

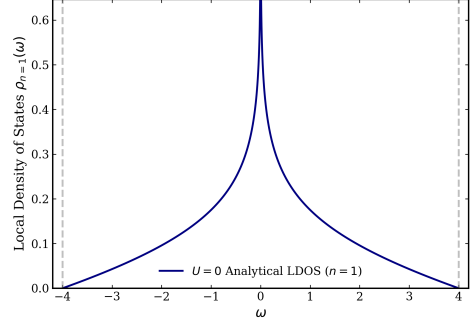


Figure 5: Analytical form of the local density of states $\rho_{n=1}(\omega)$

6 Bound State in the Fermionic Chain: Properties

6.1 Quantum Statistics of the Doublon

Since the bound state (doublon) is a composite particle made of two fermions, let us investigate its quantum statistics. To do this, we define the creation operator for a nearest-neighbor pair at site j as:

$$b_j^\dagger = c_j^\dagger c_{j+1}^\dagger \quad \text{and} \quad b_j = c_{j+1} c_j \quad (71)$$

To determine whether these composite particles behave as bosons or fermions, we will study the commutation or anticommutation relations of the b_j operators.

$$\begin{aligned}b_i b_j &= c_{i+1} c_i c_{j+1} c_j = (-1)^2 c_{j+1} c_{i+1} c_i c_j = c_{j+1} c_j c_{i+1} c_i = b_j b_i \\ b_i^\dagger b_j^\dagger &= c_i^\dagger c_{i+1}^\dagger c_j^\dagger c_{j+1}^\dagger = (-1)^4 c_j^\dagger c_{j+1}^\dagger c_i^\dagger c_{i+1}^\dagger = b_j^\dagger b_i^\dagger\end{aligned}$$

Hence, we have $[b_i, b_j] = 0$ and $[b_i^\dagger, b_j^\dagger] = 0$, which have been derived using the anticommutation relations of the underlying fermions. Now, let us evaluate the commutator $[b_i, b_j^\dagger]$.

$$\begin{aligned}b_i b_j^\dagger &= c_{i+1} c_i c_j^\dagger c_{j+1}^\dagger = c_{i+1} (\delta_{ij} - c_j^\dagger c_i) c_{j+1}^\dagger = \delta_{ij} c_{i+1} c_{j+1}^\dagger - c_{i+1} c_j^\dagger c_i c_{j+1}^\dagger \\ &= \delta_{ij} c_{i+1} c_{i+1}^\dagger - (\delta_{i+1, j} - c_j^\dagger c_{i+1}) (\delta_{i, j+1} - c_{j+1}^\dagger c_i) \\ &= \delta_{ij} c_{i+1} c_{i+1}^\dagger + \delta_{i+1, j} c_{i+2}^\dagger c_i + \delta_{i, j+1} c_j^\dagger c_{j+2} - c_j^\dagger c_{i+1} c_{j+1}^\dagger c_i\end{aligned}$$

$$\begin{aligned}
&= \delta_{ij}(1 - n_{i+1}) + \delta_{i+1,j}c_{i+2}^\dagger c_i + \delta_{i,j+1}c_j^\dagger c_{j+2} - c_j^\dagger(\delta_{i,j} - c_{j+1}^\dagger c_{i+1})c_i \\
&= \delta_{ij}(1 - n_{i+1} - n_i) + \delta_{i+1,j}c_{i+2}^\dagger c_i + \delta_{i,j+1}c_j^\dagger c_{j+2} + c_j^\dagger c_{j+1}^\dagger c_{i+1} c_i \\
\implies [b_i, b_j^\dagger] &= \delta_{ij}(1 - n_{i+1} - n_i) + \delta_{i+1,j}c_{i+2}^\dagger c_i + \delta_{i,j+1}c_j^\dagger c_{j+2}
\end{aligned}$$

Unlike bosons, which strictly obey $[b_i, b_j^\dagger] = \delta_{ij}$, the composite pairs are governed by the Pauli exclusion principle of their constituent fermions. We can understand this modified algebra by looking at three distinct spatial regimes:

1. Separated Pairs ($|i - j| > 1$): When the creation and annihilation operators act on distant, non-overlapping sites, the Kronecker deltas all vanish, yielding $[b_i, b_j^\dagger] = 0$. At long distances, the internal fermionic structure is hidden and the doublons behave entirely as pure, non-interacting bosons.

2. On-Site Commutator ($i = j$): When evaluating the pair on the exact same site, we obtain $[b_i, b_i^\dagger] = 1 - n_i - n_{i+1}$. This defines an algebra which is quite similar to that of **hard-core bosons**[10]. In a vacuum state ($\langle n_i \rangle = \langle n_{i+1} \rangle = 0$), the commutator is 1, mimicking a true boson. However, the density terms enforce an infinite on-site repulsion. If either $n_i = 1$ or $n_{i+1} = 1$, then the action of $b_i^\dagger$ kills the state, by Pauli exclusion. Since one of the sites is unoccupied, the action of $b_i = c_{i+1}c_i$ also annihilates the state. Thus, the commutator vanishes. If both $n_i = 1$ and $n_{i+1} = 1$, the action of $b_i^\dagger$ annihilates the state by Pauli exclusion. However, the action of b_i leads to the removal of the pair, yielding vacuum, on which $b_i^\dagger$ acts to give back the original state. Thus, the commutator is -1 in this case.

3. Nearest-Neighbor Overlap ($|i - j| = 1$): In this case, setting $j = i + 1$, the commutator yields $[b_i, b_{i+1}^\dagger] = c_{i+2}^\dagger c_i$. This is a single-fermion hopping term. We can understand this by looking at the action of the operators on the three involved sites, $(i, i + 1, i + 2)$:

If the shared middle site is empty (e.g., the state $|1, 0, 0\rangle$), the action of $b_i b_{i+1}^\dagger$ first creates a pair at $(i + 1, i + 2)$, momentarily forming a filled cluster $|1, 1, 1\rangle$. Then, b_i annihilates the pair at $(i, i + 1)$, leaving behind a single fermion at site $i + 2$ (the state $|0, 0, 1\rangle$). The reverse operation, $b_{i+1}^\dagger b_i$, yields 0 because there is no initial pair at $(i, i + 1)$ to annihilate. Thus, the commutator yields a net transfer of a fermion from i to $i + 2$.

If the shared middle site is occupied (e.g., the state $|1, 1, 0\rangle$), the action of $b_{i+1}^\dagger$ is immediately blocked by Pauli exclusion since site $i + 1$ is already filled; thus $b_i b_{i+1}^\dagger$ yields 0. Conversely, $b_{i+1}^\dagger b_i$ first annihilates the pair at $(i, i + 1)$ leaving the vacuum $|0, 0, 0\rangle$, and then creates a pair at $(i + 1, i + 2)$, resulting in the state $|0, 1, 1\rangle$. Subtracting this yields $-|0, 1, 1\rangle$, perfectly matching the fermionic minus sign acquired by the single-particle operator $c_{i+2}^\dagger c_i$.

In both cases, the net action is the transfer of a constituent fermion from one doublon to the other like an effective hopping.

6.2 Effective Hamiltonian for the Bound State

In the previous sections, we have established the existence of a two-particle bound state in the fermionic chain in the strong coupling regime ($|U| \gg t$). To understand the dynamics of this bound state, we can derive an effective Hamiltonian that describes the motion of the doublon as a composite particle.

6.2.1 The Feshbach Projection Formalism

The Feshbach projection method is a technique that allows us to derive an effective Hamiltonian for a specific subspace of interest by integrating out the degrees of freedom associated with the complementary subspace[11, 12].

Let us begin by partitioning the total Hilbert space into two mutually orthogonal sectors using projection operators P and Q :

- **The P -space:** The 2-fermion subspace spanned by the nearest-neighbor pair configurations, which form the bound states (doublons).
- **The Q -space:** The complementary subspace containing all other two-fermion configurations.

By definition, these projectors satisfy completeness ($P + Q = \mathbb{I}$) and orthogonality ($PQ = 0$). The total Hamiltonian $\hat{H}$ can be decomposed into block matrices acting within and between these spaces:

$$\hat{H} = \begin{pmatrix} H_{PP} & H_{PQ} \\ H_{QP} & H_{QQ} \end{pmatrix} \quad (72)$$

where, for instance, $H_{PQ} = P\hat{H}Q$ represents the operator that transitions a state from the Q -space back into the bound P -space.

Our goal is to solve the exact stationary Schrödinger equation, $\hat{H}|\psi\rangle = \omega|\psi\rangle$. By decomposing the full eigenstate into its respective projections, $|\psi\rangle = (|\psi_P\rangle, |\psi_Q\rangle)$, we can project the Schrödinger equation into the two subspaces. This yields a system of coupled equations:

$$H_{PP}|\psi_P\rangle + H_{PQ}|\psi_Q\rangle = \omega|\psi_P\rangle \quad (73)$$

$$H_{QP}|\psi_P\rangle + H_{QQ}|\psi_Q\rangle = \omega|\psi_Q\rangle \quad (74)$$

From the second equation, we can isolate $|\psi_Q\rangle$:

$$|\psi_Q\rangle = (\omega - H_{QQ})^{-1}H_{QP}|\psi_P\rangle \quad (75)$$

Here, the operator $(\omega - H_{QQ})^{-1}$ is the resolvent of the Q -space.

Substituting this expression back into the first equation yields a closed-form eigenvalue problem defined entirely within the bound-state P -space:

$$[H_{PP} + H_{PQ}(\omega - H_{QQ})^{-1}H_{QP}]|\psi_P\rangle = \omega|\psi_P\rangle \quad (76)$$

The term in the brackets acting on $|\psi_P\rangle$ is the exact, energy-dependent **effective Hamiltonian**, $\hat{H}_{\text{eff}}(\omega)$ [13]:

$$\hat{H}_{\text{eff}}(\omega) = H_{PP} + H_{PQ}(\omega - H_{QQ})^{-1}H_{QP} \quad (77)$$

This result encodes the physical narrative of the effective dynamics into distinct mathematical operators:

1. H_{PP} : The bare energy of the bound state.
2. H_{QP} : The breaking of the pair into the high-energy continuum.
3. $(\omega - H_{QQ})^{-1}$: The virtual propagation of the separated fermions.
4. H_{PQ} : The recombination of the fermions into a shifted nearest-neighbor bound state.

Alternatively, we can also derive the effective Hamiltonian using the technique of matrix inversion. Starting from the full Hamiltonian, we can write the resolvent as:

$$G = (\omega - \hat{H})^{-1} = \begin{pmatrix} \omega - H_{PP} & -H_{PQ} \\ -H_{QP} & \omega - H_{QQ} \end{pmatrix}^{-1} = \begin{pmatrix} G_{PP} & G_{PQ} \\ G_{QP} & G_{QQ} \end{pmatrix} \quad (78)$$

Using the formula for the inverse of a block matrix, we find that $G_{PP} = (\omega - H_{PP} - H_{PQ}(\omega - H_{QQ})^{-1}H_{QP})^{-1}$. Now, we can identify the effective Hamiltonian as $G_{PP} = (\omega - \hat{H}_{\text{eff}})^{-1}$, which leads us to:

$$\hat{H}_{\text{eff}}(\omega) = H_{PP} + H_{PQ}(\omega - H_{QQ})^{-1}H_{QP} \quad (79)$$

This is the same result as obtained from the projection method, but the projection method provides a more rigorous approach to derive the effective Hamiltonian.

6.2.2 Block Structure of the Hamiltonian

To apply the Feshbach formalism, we must explicitly construct the block operators H_{PP} , H_{QP} , H_{PQ} , and H_{QQ} for our specific model. We consider the total Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}$, where the unperturbed part captures the nearest-neighbor interactions and the kinetic hopping is treated as a perturbation:

$$\hat{H}_0 = U \sum_j n_j n_{j+1} ; \quad \hat{V} = -t \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) \quad (80)$$

Conventionally, we treat the kinetic term as the unperturbed Hamiltonian and the interaction as the perturbation. However, in the strong coupling regime ($|U| \gg t$), the U term dominates and the kinetic term can be treated as a perturbation. Also, the bound state is an eigenstate of the interaction term, so in studying bound state dynamics, this choice becomes more natural.

The P -space is spanned by the basis states $|B_j\rangle = c_j^\dagger c_{j+1}^\dagger |0\rangle$. The corresponding projection operator is $P = \sum_j |B_j\rangle\langle B_j|$. The orthogonal Q -space containing all other configurations is defined by the projector $Q = \mathbb{I} - P$.

The H_{PP} block describes the unperturbed dynamics strictly inside the bound-state subspace: $H_{PP} = P(\hat{H}_0 + \hat{V})P$. When the kinetic operator $\hat{V}$ acts on a bound state $|B_j\rangle$, it moves one fermion by a single site, resulting in a configuration in the Q -space. As a result, projecting it back into the P -space yields exactly zero ($P\hat{V}P = 0$). Conversely, the interaction operator simply counts the pairs. Since every state in P possesses exactly one nearest-neighbor pair, $\hat{H}_0|B_j\rangle = U|B_j\rangle$. Thus:

$$H_{PP} = U \sum_j |B_j\rangle\langle B_j| \quad (81)$$

This H_{QP} block dictates the transition from the bound state into the broken-pair continuum: $H_{QP} = Q(\hat{H}_0 + \hat{V})P$. As established, $\hat{H}_0$ acting on a P -state returns a P -state, which vanishes upon Q -projection ($Q\hat{H}_0P = 0$).

Evaluating the action of $\hat{V}$ on $|B_j\rangle$ yields two possible intermediate pair-breaking hops. Let us define $M_j = c_j^\dagger c_{j+2}^\dagger |0\rangle$ as the state where the two fermions are separated by one empty site. Then, the two hoppings are:

$$-t(c_{j-1}^\dagger c_j) c_j^\dagger c_{j+1}^\dagger |0\rangle = -t c_{j-1}^\dagger c_{j+1}^\dagger |0\rangle = -t |M_{j-1}\rangle \quad (82)$$

$$-t(c_{j+2}^\dagger c_{j+1}) c_j^\dagger c_{j+1}^\dagger |0\rangle = t c_{j+2}^\dagger c_j^\dagger |0\rangle = -t c_{j+2}^\dagger c_{j+2}^\dagger |0\rangle = -t |M_j\rangle \quad (83)$$

Summing over all sites:

$$H_{QP} = -t \sum_j \left(|M_{j-1}\rangle\langle B_j| + |M_j\rangle\langle B_j| \right) \quad (84)$$

Because the Hamiltonian is Hermitian,

$$H_{PQ} = H_{QP}^\dagger = -t \sum_j \left(|B_j\rangle\langle M_{j-1}| + |B_j\rangle\langle M_j| \right) \quad (85)$$

The final block, $H_{QQ} = Q(\hat{H}_0 + \hat{V})Q$, describes the dynamics within the Q -space. Since the Q -space contains all configurations except for the nearest-neighbor pairs, the interaction term $\hat{H}_0$ vanishes on any state in Q and thus $H_{QQ} = Q\hat{V}Q$. Finding a closed form expression for H_{QQ} is nontrivial due to the large number of configurations in the Q -space, but we won't need it for our analysis.

This is because, all we need to compute the effective Hamiltonian is the resolvent $(\omega - H_{QQ})^{-1}$, which can be approximated in the $\omega = U \gg t$ limit.

$$(\omega - H_{QQ})^{-1} = (U - Q\hat{V}Q)^{-1} = \frac{1}{U} \left(1 - \frac{Q\hat{V}Q}{U} \right)^{-1} \quad (86)$$

Note that the operator $Q\hat{V}Q$ is of order t , and thus the term $\frac{Q\hat{V}Q}{U}$ is of order $\frac{t}{U} \ll 1$.

This allows one to expand the inverse operator in a power series and truncate it at the zeroth order in $\frac{t}{U}$: So, the resolvent simplifies to

$$(\omega - H_{QQ})^{-1} \approx \frac{1}{U} \mathbb{I}_Q \quad (87)$$

6.2.3 Effective Hamiltonian

With the block matrices, we can now evaluate the Feshbach formula for the effective Hamiltonian of the bound state subspace:

$$\hat{H}_{\text{eff}} = H_{PP} + H_{PQ}(\omega - H_{QQ})^{-1}H_{QP} \quad (88)$$

Using the zeroth-order approximation for the resolvent, the second term becomes:

$$\begin{aligned} H_{PQ} \left(\frac{1}{U} \right) H_{QP} &= \frac{1}{U} \left[-t \sum_i (|B_i\rangle\langle M_{i-1}| + |B_i\rangle\langle M_i|) \right] \left[-t \sum_j (|M_{j-1}\rangle\langle B_j| + |M_j\rangle\langle B_j|) \right] \\ &= \frac{t^2}{U} \sum_{i,j} |B_i\rangle \left(\langle M_{i-1}|M_{j-1}\rangle + \langle M_{i-1}|M_j\rangle + \langle M_i|M_{j-1}\rangle + \langle M_i|M_j\rangle \right) \langle B_j| \end{aligned} \quad (89)$$

Because the $|M_k\rangle$ states represent distinct spatial configurations of the separated fermions, their inner products yield Kronecker deltas: $\langle M_k|M_l\rangle = \delta_{k,l}$. This leads to the following simplifications:

$$\begin{aligned} H_{PQ} \left(\frac{1}{U} \right) H_{QP} &= \frac{t^2}{U} \sum_i (|B_i\rangle\langle B_i| + |B_i\rangle\langle B_{i-1}| + |B_i\rangle\langle B_{i+1}| + |B_i\rangle\langle B_i|) \\ &= \sum_i \left[\frac{2t^2}{U} |B_i\rangle\langle B_i| + \frac{t^2}{U} (|B_i\rangle\langle B_{i-1}| + |B_i\rangle\langle B_{i+1}|) \right] \end{aligned} \quad (90)$$

Adding the first term, $H_{PP} = U \sum_i |B_i\rangle\langle B_i|$, we obtain the effective Hamiltonian for the doublon as:

$$\hat{H}_{\text{eff}} = \sum_j \left(U + \frac{2t^2}{U} \right) |B_j\rangle\langle B_j| + \frac{t^2}{U} \sum_j (|B_j\rangle\langle B_{j-1}| + |B_j\rangle\langle B_{j+1}|) \quad (91)$$

This result demonstrated the following physical picture - in the strong coupling regime, the tightly bound pair of fermions behaves collectively as a single, composite particle on a lattice[14]. This composite doublon is governed by a standard tight-binding model, but with renormalized parameters: its effective on-site potential energy is shifted to $\tilde{U} = U + 2t^2/U$, and it acquires an effective nearest-neighbor hopping amplitude of $J_{\text{eff}} = t^2/U$. The reduced hopping rate ($J_{\text{eff}} \ll t$) mathematically reflects the fact that the doublon can only move via a slow process of breaking and recombining.

6.3 Local Density of States of the Bound State

Having mapped the dynamics of the composite doublon to a standard tight-binding model via the Feshbach formalism, we can now deduce the local density of states (LDOS) for the nearest-neighbor sector, which we previously computed numerically.

To calculate the width and boundaries of this emergent band, we diagonalize the effective Hamiltonian. We introduce the Bloch states for the center-of-mass momentum K of the doublon:

$$|K\rangle = \frac{1}{\sqrt{N}} \sum_j e^{iKja} |B_j\rangle \quad (92)$$

Applying $\hat{H}_{\text{eff}}$ to these momentum eigenstates yields the energy dispersion relation for the composite particle:

$$\omega_{BS}(K) = \tilde{U} + 2J_{\text{eff}} \cos(Ka) = \left(U + \frac{2t^2}{U} \right) + \frac{2t^2}{U} \cos(Ka) \quad (93)$$

The width of this bound-state band, W_{BS} , is given by the difference between the maximum and minimum energies of the dispersion relation (which occur at $K = 0$ and $K = \pi/a$):

$$W_{BS} = \omega_{\text{max}} - \omega_{\text{min}} = 4|J_{\text{eff}}| = \frac{4t^2}{|U|} \quad (94)$$

The scattering continuum possesses a bandwidth of $W_{\text{cont}} = 8t$. In contrast, the bandwidth of the composite doublon scales inversely with U . The ratio of their widths is proportional to $t/U \ll 1$, meaning the doublon band is extremely narrow.

Using the standard result for the density of states of a 1D dispersion, the exact analytical form of the local density of states for the doublon band is:

$$\rho_{BS}(\omega) = \frac{1}{\pi \sqrt{4J_{\text{eff}}^2 - (\omega - \tilde{U})^2}} \quad \text{for } |\omega - \tilde{U}| < 2|J_{\text{eff}}| \quad (95)$$

This confirms that the doublon manifests as an isolated, narrow band separated from the scattering continuum, which diverges at the edges $\tilde{U} \pm 2|J_{\text{eff}}|$ and vanishes outside this range. As shown in figure 6, this analytical form perfectly matches the numerical local density of states for the bound state we previously computed, confirming the validity of our effective Hamiltonian description.

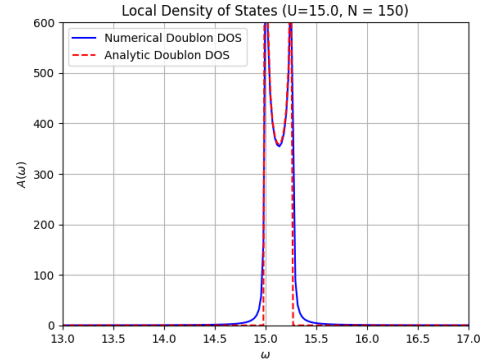


Figure 6: Comparison of the local density of states for the bound state with the analytical form derived from the effective Hamiltonian.

7 Conclusion

In this report, we have investigated the formation, spectral signatures, and emergent dynamics of two-particle bound states in a one-dimensional interacting fermionic chain. Motivated by the exact Jordan-Wigner equivalence to the Heisenberg spin chain, this study provides a microscopic perspective on how strongly correlated elementary excitations - such as magnons - can bind together to form stable composite quasiparticles.

By calculating the two-particle Green's function, we explicitly determined the criteria for bound state formation. We demonstrated that a normalizable bound state strictly emerges when the interaction strength exceeds the kinetic energy scale ($|U| > 2t$). Through the exact analytical derivation of the spectral weights and the local density of states, we mapped the bound state as it physically separates

from the two-particle scattering continuum. In the non-interacting limit, the density of states features the broad continuous band characteristic of independent fermions. However, as the system enters the strong-coupling regime ($|U| \gg t$), the spectral weight collapses into a narrow energy band, representing a highly localized “doublon”.

To capture the dynamics of this doublon, we applied the Feshbach projection operator formalism to integrate out the high-energy broken-pair continuum. This procedure revealed that the doublon behaves exactly as a single composite particle governed by a renormalized tight-binding Hamiltonian. Because the pair can only move through a process of breaking and recombining, its effective hopping amplitude is severely suppressed to $J_{\text{eff}} = t^2/U$. This perfectly explains the extreme narrowing of the bound-state density of states and solidifies the picture of bound states as heavy, slow-moving quasiparticles.

The emergence of such heavy composite particles has profound implications for the non-equilibrium dynamics of strongly correlated systems[14]. Because bound states move much slower than isolated excitations and require massive energy exchanges to break apart, they can act as kinetic bottlenecks[15]. This phenomenon is intimately tied to the slow thermalization, and robust coherence observed in modern quantum simulator experiments using ultracold atoms and superconducting qubits[16, 17].

Future extensions of this work could explore larger multi-particle bound states, the scattering cross-sections between interacting doublons, or the fate of these bound states upon the introduction of impurities and lattice disorder. Ultimately, the framework presented in this report highlights the significance of emergent phenomena in quantum many-body physics: how a simple competition between localized interactions and kinetic hopping can give birth to an entirely new class of quasiparticles.

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